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1.1 Water & Carbon Cycles

1. Systems Concepts

System: A set of interrelated components working together to form a functioning whole. In geography, the water and carbon cycles are examples of *closed systems* at a global scale (no matter enters or leaves, though energy does) and *open systems* at a local scale (both energy and matter cross the boundary).

Key systems terminology:

- **Inputs** — energy or matter entering the system (e.g. precipitation entering a drainage basin, solar radiation driving evaporation)
- **Outputs** — energy or matter leaving the system (e.g. river discharge to the ocean, evapotranspiration to the atmosphere)
- **Stores (components)** — where energy or matter is held (e.g. soil moisture, aquifers, biomass, ocean water, permafrost carbon)
- **Flows (transfers/fluxes)** — movement of energy or matter between stores (e.g. infiltration, throughflow, photosynthesis, respiration)
- **Positive feedback** — amplifies change, pushing the system further from equilibrium (e.g. ice-albedo effect: warming melts ice, reducing reflectivity, causing more warming)
- **Negative feedback** — dampens change, restoring equilibrium (e.g. increased evaporation leads to more cloud cover, reflecting solar radiation and reducing temperature)
- **Dynamic equilibrium** — a state of balance maintained despite ongoing fluctuations; the system self-regulates around an average condition
- **Threshold** — a critical point beyond which a system shifts to a new state of equilibrium (e.g. permafrost thawing irreversibly once temperatures exceed a threshold)

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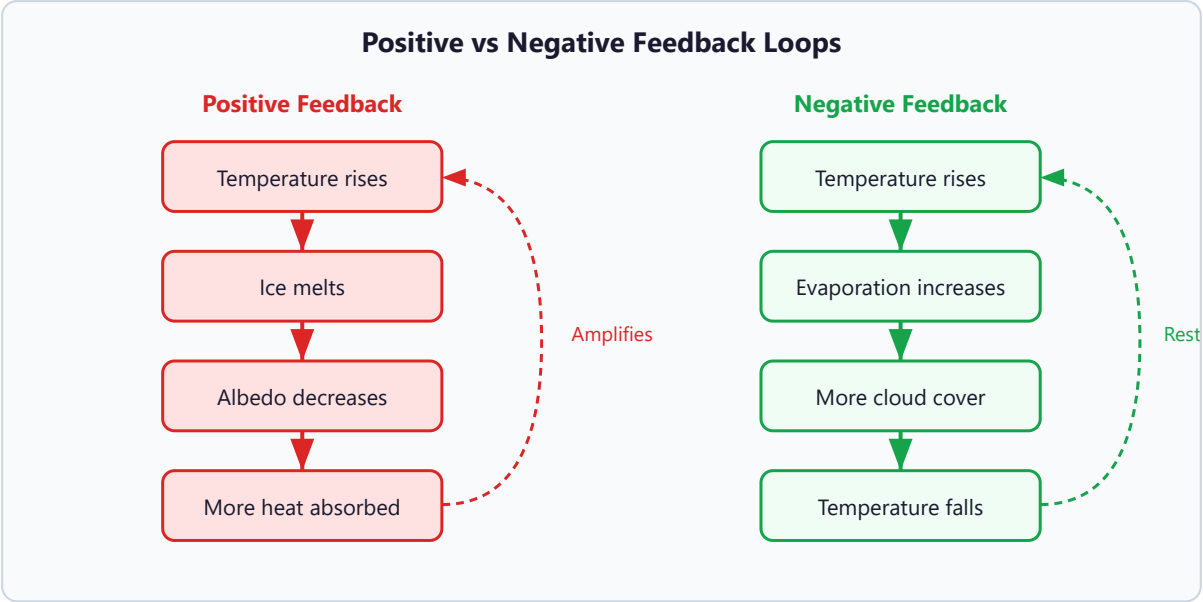

2. The Global Water Cycle

Hydrological cycle: The continuous movement of water between the atmosphere, land, and oceans through evaporation, condensation, precipitation, and runoff. At a global scale it is a *closed system* — no water enters or leaves the Earth system, though solar energy drives the cycle. The total volume of water on Earth is approximately 1.386 billion km³.

Major Stores of Water

Store	Volume (10^3 km^3)	% of Total	Residence Time
Oceans	1,335,040	96.5%	~3,600 years
Glaciers & ice caps	26,350	1.74%	~10,000 years (up to 1 million years in Antarctica)
Groundwater (total)	10,530	0.76%	100–10,000 years
Permafrost	300	0.022%	10,000+ years
Freshwater lakes	91	0.007%	~17 years
Soil moisture	16.5	0.001%	1–2 months

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Atmosphere	12.9	0.001%	~10 days
Rivers	2.12	0.0002%	~2 weeks
Biosphere	1.12	0.0001%	~1 week

Only ~2.5% of all water is freshwater, and of that, ~68.7% is locked in ice caps and glaciers. Only ~0.3% of all freshwater is in surface stores (rivers and lakes) readily available for human use.

Key Fluxes (Transfers)

- **Evaporation** — liquid water converts to vapour, driven by solar energy. ~505,000 km^3/yr evaporates globally, with ~86% from ocean surfaces. Rate increases with temperature, wind speed, and low humidity.
- **Transpiration** — water loss through plant stomata during photosynthesis. Tropical rainforests are major contributors: the Amazon alone returns ~8 trillion litres/day to the atmosphere.
- **Evapotranspiration** — combined water loss from evaporation and transpiration. Potential evapotranspiration (PET) is the amount that would occur with unlimited water supply; actual evapotranspiration depends on available moisture.
- **Condensation** — vapour cools to dew point, forming water droplets/ice crystals on condensation nuclei, creating clouds.
- **Precipitation** — water falling from clouds as rain, snow, sleet, or hail. Global average is ~990 mm/yr over land. Three main types: convectional, orographic (relief), and frontal.
- **Interception** — precipitation caught by vegetation or buildings before reaching the ground. Can account for 10–40% of rainfall in forested areas.
- **Stemflow** — water running down plant stems and trunks to the ground.
- **Infiltration** — water soaking into the soil from the surface. Rate depends on soil type, antecedent moisture, vegetation cover, and land use.
- **Percolation** — deeper downward movement of water through soil into bedrock and aquifers.
- **Throughflow** — lateral movement of water through the soil layer, following the gradient.
- **Groundwater flow** — slow lateral movement of water through permeable rock (aquifers). Very slow: metres per year in some aquifers.

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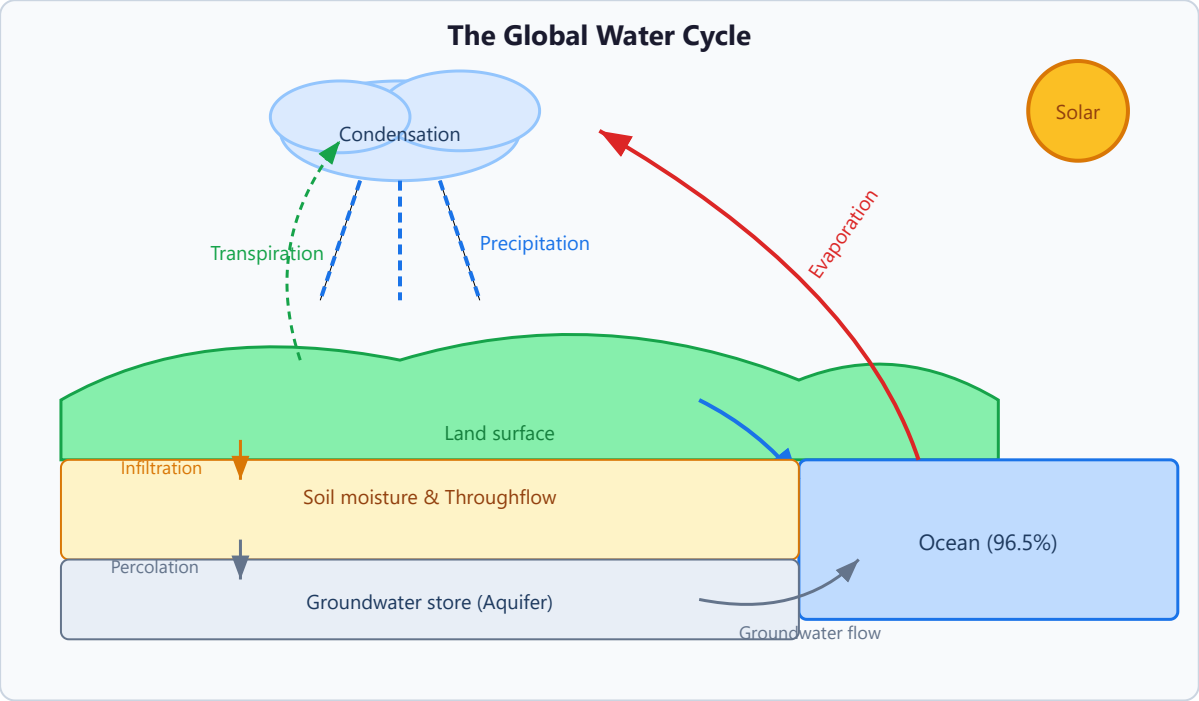
- **Surface runoff (overland flow)** — water flowing over the surface when rainfall exceeds infiltration capacity, or when the soil is saturated. Fastest route to river channels.
- **Channel flow** — water flowing within a defined river channel towards the sea.
- **Cryospheric processes** — accumulation (snowfall adding to glaciers), ablation (melting and sublimation), and glacial meltwater runoff.



Changes to the Water Cycle Over Time

The water cycle is not static. Both natural and human factors alter the distribution and movement of water:

- **Climate change** — global warming intensifies the water cycle: warmer air holds ~7% more moisture per 1°C rise (Clausius-Clapeyron relation), leading to more intense rainfall events but also longer droughts in some regions. Glacier and ice sheet melt transfers water from cryospheric stores to oceanic stores.
- **Deforestation** — removes interception and transpiration, increasing surface runoff by 20–40%, reducing soil infiltration, and lowering local humidity and rainfall. Deforestation in the Amazon has reduced regional rainfall by up to 25% in some areas.
- **Urbanisation** — impermeable surfaces reduce infiltration and groundwater recharge, increasing surface runoff volumes by 2–6 times compared to natural surfaces.
- **Abstraction** — excessive groundwater extraction depletes aquifers faster than recharge. The Ogallala Aquifer (USA) has declined by over 30 m in parts of Texas and Kansas since 1950.

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- **Agricultural irrigation** — diverts river water and depletes aquifers. The Aral Sea shrank to just 10% of its 1960 size due to irrigation abstraction from the Amu Darya and Syr Darya rivers.

3. The Drainage Basin as an Open System

Drainage basin: The area of land drained by a river and its tributaries. It is an *open system* with inputs (precipitation) and outputs (discharge, evapotranspiration). The boundary is the **watershed** — a ridge of high land separating adjacent drainage basins.

The **water balance equation** summarises the relationship between inputs and outputs:

$$P = Q + E \pm \Delta S$$

Where: P = precipitation, Q = discharge (runoff), E = evapotranspiration, ΔS = change in storage (soil moisture, groundwater, snow/ice).

In the UK, winter months typically have a **water surplus** (P > E, soil moisture recharge, rising groundwater), while summer months have a **water deficit** (E > P, drawing down soil moisture). The balance varies spatially: NW England receives >2,000 mm/yr precipitation; SE England receives <600 mm/yr.

Factors Affecting Water Balance

- **Climate** — seasonal variations in rainfall and temperature affect inputs and evapotranspiration rates.
- **Rock type (geology)** — permeable rocks (chalk, limestone, sandstone) allow infiltration and sustain baseflow; impermeable rocks (clay, granite, slate) promote surface runoff and flashier rivers.
- **Soil type** — sandy soils drain quickly with high infiltration rates (up to 25 mm/hr); clay soils hold water and generate more surface runoff (infiltration as low as 1 mm/hr).
- **Vegetation** — intercepts precipitation (10–40% in forests), increases infiltration through root channels, promotes transpiration, and binds soil to reduce erosion.
- **Land use** — urbanisation creates impermeable surfaces, increasing surface runoff. Agriculture may compact soil, reducing infiltration. Reservoirs store water and regulate discharge.
- **Relief** — steep slopes promote rapid surface runoff and shorter lag times; flat land encourages infiltration and ponding.

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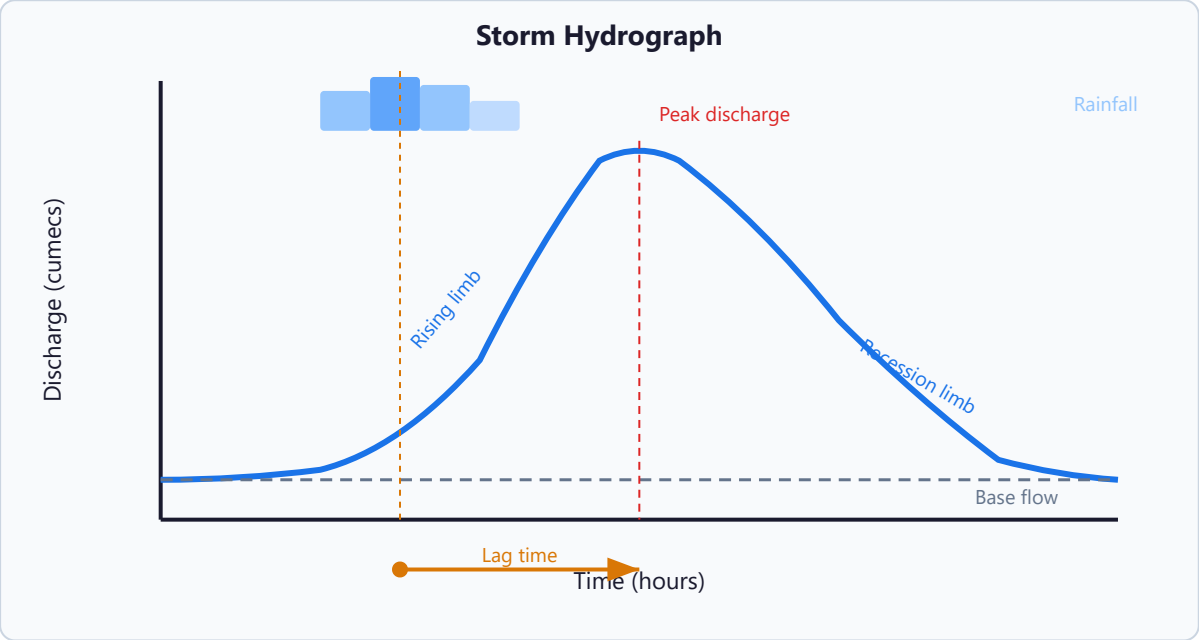
The Storm Hydrograph

Storm hydrograph: A graph showing how a river's discharge (cumecs, m^3/s) changes over time in response to a rainfall event. Key features: peak discharge, lag time (delay between peak rainfall and peak discharge), rising limb, falling (recession) limb, and baseflow.



Factors Affecting the Hydrograph Shape

Factor	Flashy Hydrograph (short lag, high peak)	Flat Hydrograph (long lag, low peak)
Basin size	Small basin — water reaches channel quickly	Large basin — longer travel distance
Basin shape	Circular — all tributaries arrive at once	Elongated — tributaries arrive sequentially
Drainage density	High density — many channels, rapid transfer	Low density — fewer channels, slower transfer
Slope/relief	Steep slopes — rapid surface runoff	Gentle slopes — slower flow, more infiltration

Rock type	Impermeable (clay, granite) — high runoff	Permeable (chalk, limestone) — high infiltration
Soil type	Saturated/clay soils — low infiltration	Sandy/dry soils — high infiltration
Land use	Urban (impermeable surfaces, drains)	Forested (interception, infiltration, transpiration)
Antecedent conditions	Saturated from previous rain	Dry conditions — soil absorbs more water
Precipitation type	Intense, short-duration storms	Prolonged, gentle rainfall or snow

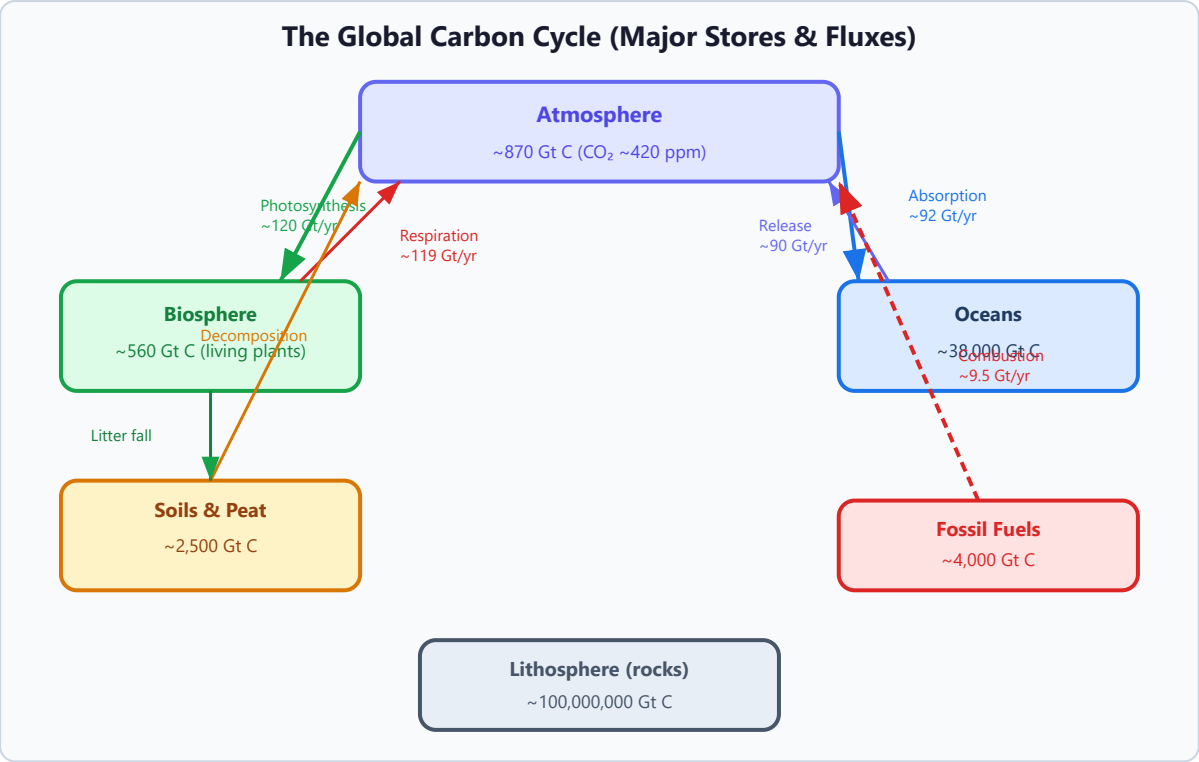
Case Study: The Boscastle Flash Flood, Cornwall (16 August 2004)
Boscastle received 185 mm of rainfall in 5 hours (a 1-in-400-year event). The River Valency and River Jordan rapidly flooded. Flashy hydrograph factors: small, steep catchment (only 23 km^2), impermeable slate bedrock, saturated soils from previous wet weather, steep valley sides funnelling water to a narrow channel. Impacts: 58 properties flooded, 115 vehicles swept away, 4 buildings demolished, estimated £30 million damage, but no deaths (due to rapid helicopter rescue of 100+ people). The peak discharge was estimated at 180 m^3/s, compared to a normal flow of <1 m^3/s. Response: £10 million flood defence scheme completed in 2008 (widened and deepened channel, raised bridge, upstream culvert improvements).

4. The Global Carbon Cycle

Carbon cycle: The biogeochemical cycle by which carbon is exchanged between the atmosphere, biosphere, hydrosphere, lithosphere, and cryosphere. Carbon is the building block of all organic life and a key component of greenhouse gases (CO_2 and CH_4). Like the water cycle, the carbon cycle is a closed system at the global scale.

Major Carbon Stores

Store	Estimated Carbon (Gt C)	Residence Time	Notes
Lithosphere (sedimentary rocks, fossil fuels)	~100,000,000	Millions of years	By far the largest store. Includes limestone ($CaCO_3$), coal, oil, and gas. Very slow cycling.
Oceans (dissolved & organic)	~38,000	Hundreds to thousands of years	Deep ocean is main store. Surface ocean exchanges rapidly with atmosphere. The ocean absorbs ~25% of annual anthropogenic CO_2.
Soil organic matter	~2,500	Decades to millennia	Includes peat, humus. The top 1 m of soil contains more carbon than the atmosphere and all vegetation combined.
Permafrost	~1,700	Thousands of years	Frozen organic matter. Thawing releases CO_2 and CH_4 — a major positive feedback risk.
Atmosphere	~870 (at ~420 ppm CO_2)	~5 years (for individual molecules)	Pre-industrial level was ~590 Gt C (280 ppm). Now ~420 ppm due to fossil fuel burning and deforestation.
Terrestrial biosphere (living plants)	~560	Years to decades	Forests are the largest biomass store. Tropical forests hold ~55% of terrestrial biomass carbon.
Fossil fuels (recoverable)	~4,000	Millions of years to form	Coal (~65%), oil (~15%), gas (~20%). Non-renewable on human timescales.



Key Carbon Fluxes

- **Photosynthesis** — plants absorb ~120 Gt C/yr from the atmosphere: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$. The largest natural carbon flux from atmosphere to biosphere.
- **Respiration** — plants, animals, and microorganisms release CO_2 back to the atmosphere (~119 Gt C/yr). Carbon stored in organic matter is returned as organisms metabolise.
- **Decomposition** — bacteria and fungi break down dead organic matter, releasing CO_2 (aerobic) or CH_4 (anaerobic, e.g. in waterlogged conditions). Rate depends on temperature, moisture, and oxygen availability.
- **Combustion** — burning fossil fuels releases ~9.5 Gt C/yr (2023 estimate); burning biomass (forest fires, slash-and-burn) releases additional carbon. This is the primary driver of rising atmospheric CO_2.
- **Ocean-atmosphere exchange** — CO_2 dissolves in surface waters (~92 Gt C/yr absorbed, ~90 Gt C/yr released). Cold water absorbs more CO_2 (solubility pump). The ocean has absorbed ~30% of anthropogenic emissions since 1750.
- **Biological pump** — marine organisms (phytoplankton) photosynthesise, incorporating carbon. When they die, carbon sinks to the deep ocean in organic matter. Crucial for long-term oceanic carbon storage.

- **Weathering** — chemical weathering of silicate and carbonate rocks by carbonic acid ($CO_2 + H_2O \rightarrow H_2CO_3$) removes atmospheric CO_2 very slowly. The weathered carbonates are transported by rivers to oceans.
- **Sedimentation and lithification** — carbon-rich sediments accumulate on ocean floors and are eventually compressed into sedimentary rocks (limestone, chalk). The geological carbon cycle operates over millions of years.
- **Volcanic outgassing** — volcanoes release CO_2 from the lithosphere back to the atmosphere (~0.3 Gt C/yr). Small compared to anthropogenic emissions but important over geological time.

The Biological Carbon Pump and Sequestration

Carbon sequestration: The process of capturing and storing atmospheric carbon dioxide. Natural sequestration occurs through photosynthesis (forests, oceans) and soil/peat formation. Artificial sequestration includes Carbon Capture and Storage (CCS) technology, which captures CO_2 from power stations and injects it underground.

Carbon sink: A store that absorbs more carbon than it releases (net removal from atmosphere). Currently, forests and oceans are the major sinks, together absorbing ~55% of anthropogenic CO_2 emissions each year.
Carbon source: A store that releases more carbon than it absorbs (net addition to atmosphere). Fossil fuel burning, deforestation, and peatland drainage are major sources.

5. Carbon, Climate Change, and Human Impacts

The Enhanced Greenhouse Effect

The natural greenhouse effect keeps Earth ~33°C warmer than it would otherwise be. Human activities have enhanced this by increasing atmospheric concentrations of CO_2 (from ~280 ppm pre-industrial to ~420 ppm in 2024), CH_4 (from ~722 ppb to ~1,920 ppb), and N_2O. These gases trap outgoing longwave radiation, warming the troposphere.

Human Disruption of the Carbon Cycle

- **Fossil fuel combustion** — releases ~9.5 Gt C/yr, transferring carbon from geological stores (where it was sequestered over millions of years) to the atmosphere in decades. The largest single human disruption.
- **Deforestation** — releases ~1.5 Gt C/yr. Burning and decomposition of cleared trees release stored carbon; the lost forest can no longer sequester carbon. Tropical deforestation accounts for ~95% of this flux.
- **Agriculture** — rice paddies and livestock produce CH_4; fertiliser use produces N_2O; soil disturbance releases carbon. Agriculture accounts for ~10–12% of global GHG emissions.
- **Peatland drainage** — when peat is drained for farming, the exposed organic matter decomposes aerobically, releasing CO_2. Indonesia's peatland fires in 2015 released ~1.6 Gt CO_2 in just 3 months — more than Germany's annual emissions.
- **Cement production** — heating limestone ($CaCO_3$) releases CO_2. Cement accounts for ~8% of global CO_2 emissions.
- **Ocean acidification** — increased atmospheric CO_2 dissolves in oceans, forming carbonic acid. Ocean pH has fallen from 8.2 to 8.1 since 1750 (a 26% increase in acidity), threatening coral reefs and shell-forming organisms.

Case Study: The Amazon Rainforest — Carbon Store Under Threat

The Amazon is the world's largest tropical rainforest (5.5 million km^2), storing an estimated 150–200 Gt C in biomass and soils. It acts as a major carbon sink, absorbing ~0.5 Gt C/yr through photosynthesis. However, deforestation (peaking at ~27,000 km^2/yr in 2004, reduced to ~10,000 km^2/yr by 2023 under stricter enforcement) and increasing drought frequency are threatening this role. Research suggests parts of the eastern Amazon have already become a net carbon source. The 2019 fire season saw 80,000+ fires, many set deliberately for cattle ranching and soy farming. Deforestation releases stored carbon immediately (burning) and over decades (decomposition). The Amazon also generates ~50% of its own rainfall through transpiration; continued deforestation could trigger a "tipping point" (estimated at 20–25% forest loss, with ~17% already lost) where the forest dries irreversibly and converts to savanna — releasing 50+ Gt C.

Case Study: The Arctic Permafrost — Carbon Time Bomb

Arctic permafrost (permanently frozen ground) covers ~23 million km^2 and contains ~1,700 Gt C — roughly twice the carbon currently in the atmosphere. As the Arctic warms at 2–3 times the global

average rate, permafrost is thawing. This exposes previously frozen organic matter to microbial decomposition, releasing CO_2 (aerobic conditions) and CH_4 (anaerobic conditions in waterlogged areas). CH_4 is ~80 times more potent than CO_2 over 20 years. This is a classic positive feedback: warming → permafrost thaws → carbon released → more warming. Studies estimate permafrost could release 92–260 Gt C by 2100 under a high-emissions scenario. Siberian "thermokarst" lakes formed by thawing are already bubbling with methane. This feedback is largely irreversible on human timescales.

Case Study: UK Peatlands — Carbon Management

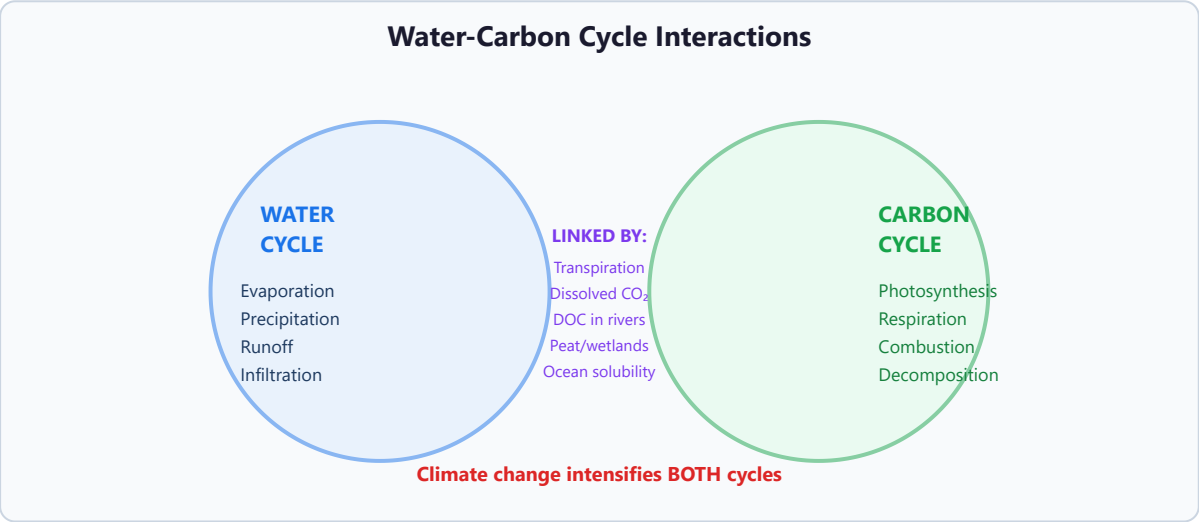
The UK has ~3 million hectares of peatland (12% of land area), storing an estimated 3.2 billion tonnes of carbon — more than all UK forests combined. However, ~80% of UK peatlands are in a degraded state due to drainage for agriculture, overgrazing, burning for grouse moor management, and peat extraction for horticulture. Degraded peatlands are a net carbon source, releasing ~23 million tonnes CO_2/yr. The UK government's 2021 Peat Action Plan aims to restore 35,000 hectares by 2025 and ban peat sales for horticulture by 2024 (delayed to 2030). Restoration involves blocking drainage ditches, rewetting, and sphagnum moss reintroduction. The Yorkshire Peat Partnership has restored 35,000+ hectares on the Pennines, reducing carbon emissions and improving water quality.

6. Water-Carbon Cycle Interactions

The water and carbon cycles are closely interconnected:

- **Photosynthesis requires water** — plants use water from soil moisture, combining it with atmospheric CO_2 to produce glucose. Drought stress reduces photosynthesis and carbon uptake.
- **Transpiration links both cycles** — water lost through stomata (water cycle flux) occurs simultaneously with CO_2 absorption (carbon cycle flux). Plants face a trade-off between water loss and carbon gain.
- **Dissolved carbon in water** — rivers transport dissolved organic carbon (DOC) and dissolved inorganic carbon (DIC) from land to oceans. The global riverine carbon flux is ~0.9 Gt C/yr.
- **CO_2 solubility in water** — CO_2 dissolves in cold ocean water, forming carbonic acid (H_2CO_3). This links the atmospheric carbon store to the oceanic carbon store. Cold polar waters are major CO_2 sinks; warm tropical waters tend to release CO_2.

- **Wetland carbon storage requires water** — waterlogged conditions create anaerobic environments where decomposition is slow, allowing peat to accumulate carbon over millennia. Draining wetlands exposes this carbon to rapid aerobic decomposition.
- **Climate change intensifies both cycles** — a warmer atmosphere holds more moisture (intensifying rainfall and evaporation) while simultaneously increasing CO_2 solubility pressure in oceans and accelerating decomposition rates on land.
- **Ocean circulation transfers both** — thermohaline circulation moves warm surface water poleward and returns cold, CO_2-rich deep water to the surface. Changes in ocean circulation (from ice melt freshening) could alter both heat distribution and carbon cycling.



7. Managing the Water and Carbon Cycles

Water Cycle Management

- **Flood management** — hard engineering (dams, levees, channel straightening) vs soft engineering (floodplain restoration, afforestation, SuDS). Pickering, North Yorkshire's "Slow the Flow" scheme used leaky dams, tree planting, and bund creation to reduce flood peaks by ~15–20%.
- **Water conservation** — metering, leak repair (UK water companies lose ~3 billion litres/day to leaks), efficient irrigation (drip irrigation uses 30–50% less water than flood irrigation).
- **Catchment management** — integrated approaches that consider the entire drainage basin. United Utilities' work in the Lake District plants native woodland to increase interception and infiltration.

Carbon Cycle Management

- **Reducing emissions** — transitioning from fossil fuels to renewables, energy efficiency, carbon taxation, international agreements (Paris Agreement: limit warming to 1.5–2°C).
- **Carbon capture and storage (CCS)** — capturing CO_2 from industrial sources and storing it in geological formations. The Sleipner project (Norway) has stored ~1 million tonnes CO_2/yr since 1996 in a saline aquifer beneath the North Sea.
- **Afforestation and reforestation** — planting trees to absorb CO_2. The UK aims to plant 30,000 hectares/yr of new woodland by 2025. A hectare of oak woodland absorbs ~6.5 tonnes CO_2/yr.
- **Peatland and wetland restoration** — rewetting degraded peatlands to restore their carbon sink function and reduce emissions.
- **Soil management** — no-till farming, cover crops, and agroforestry can increase soil carbon storage.
- **Blue carbon** — protecting and restoring coastal ecosystems (mangroves, salt marshes, seagrass beds) that sequester carbon at rates 2–4 times greater than tropical forests per unit area.

Exam Tip: AQA frequently asks about the *interconnections* between the water and carbon cycles. Always explain the link between both cycles when discussing topics like deforestation (reduces transpiration AND releases stored carbon), climate change (intensifies the water cycle AND disrupts carbon sinks), or peatland management (water table controls carbon storage). Use specific data and named examples.

Exam Tip: In 20-mark essay questions, structure your answer with a clear argument. For example, if asked whether human activity is the main cause of changes to the carbon cycle, discuss: (1) natural variations (volcanic eruptions, Milankovitch cycles, ocean circulation changes), (2) human impacts (fossil fuels, deforestation, agriculture), (3) the scale and pace of change (human impacts are unprecedented in geological history). Always evaluate which factor is most significant and justify your conclusion.

Exam Tip: Know the difference between short-term (fast) and long-term (slow) carbon cycle processes. Fast: photosynthesis, respiration, decomposition (years to decades). Slow: weathering, sedimentation, fossil fuel formation (millions of years). Human fossil fuel burning is unique because it transfers carbon from the slow cycle to the fast cycle at an unprecedented rate.

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